TAILSHARP

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The Sharpness Rating

Separating Skill from Luck in Prediction Markets

A behavioural and statistical methodology for classifying prediction-market participants and estimating persistent forecasting skill from on-chain activity.

TailSharp Research

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Abstract

Public leaderboards in prediction markets overwhelmingly rank luck. Win rate is uninformative without odds conditioning, raw return is dominated by variance at the sample sizes most wallets exhibit, and the best-performing accounts on naive metrics are frequently market makers, arbitrageurs, or manipulators rather than forecasters. This paper specifies the TailSharp Sharpness Rating: a methodology that (i) classifies on-chain prediction-market participants into behavioural types using execution-level features, (ii) decomposes observed performance into a persistent skill component and a transient luck/regime component, and (iii) estimates each wallet's forward edge with a hierarchical Bayesian model whose priors are shifted by behavioural covariates and whose published rank is a conservative lower confidence bound rather than a point estimate. We further specify the wallet-clustering and integrity layer required to resist leaderboard manipulation, and a walk-forward validation protocol under which the rating must demonstrate out-of-sample predictive power over future closing-line value. The system is an information product: it identifies and contextualises skill; it does not execute, route, or advise individual transactions.

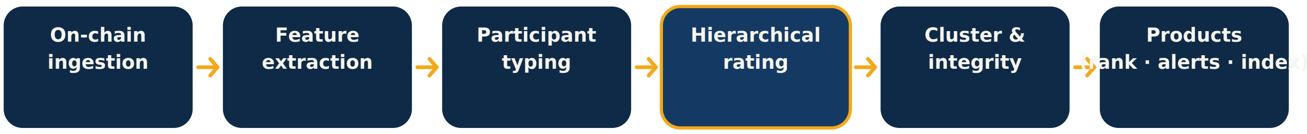

Figure 1 — The Sharpness Rating pipeline. The hierarchical rating (outlined) is the core estimator; typing and clustering protect it from non-forecasting flow and manipulation.

1 · Introduction: why leaderboards lie

Prediction markets produce a seductive illusion of measurability. Every position is public on-chain, every market resolves to a verifiable outcome, and so it appears trivial to rank participants by performance. In practice three failures make naive rankings close to worthless.

- **Odds blindness.** A binary contract bought at 95¢ wins ~95% of the time by construction. Win-rate leaderboards are therefore dominated by late-sweepers harvesting near-resolved markets, whose tail risk is severe and whose flow carries no information.
- **Sample-size blindness.** At the trade counts typical of retail wallets, the distributions of outcomes for modestly skilled and modestly unskilled participants overlap heavily. Short hot streaks are abundant in any large population by chance alone; rankings that do not shrink small samples systematically promote what we term lucky monkeys.
- **Type blindness.** The most profitable wallets on raw P&L are often not forecasters at all: market makers earning spread, arbitrageurs exploiting YES/NO pricing inconsistencies, or paired wallets wash-trading a record into existence. Following their "positions" is meaningless or worse.

The Sharpness Rating addresses all three. It conditions every measurement on price, shrinks every estimate by its evidence, and types every wallet before ranking it. The objective is precise:

publish, for each eligible wallet, a conservative estimate of *forward* edge — the component of performance expected to persist — together with the behavioural evidence supporting it.

2 · Data: price-relative accounting on-chain

The unit of observation is the fill: a wallet, a market, a side, a price, a size, and a timestamp, reconstructed from the venue's on-chain settlement layer and order-book API. Three accounting conventions govern everything downstream.

- **Digital-option framing.** Each binary position is treated as a digital option: an entry at price p is an implied probability claim, and per-trade return is measured relative to p, not in raw dollars. This makes a 55¢ entry that resolves YES and a 95¢ entry that resolves YES incomparable in exactly the way they should be.
- **Mark-to-close, not mark-to-resolution.** Wherever possible, trades are scored against the market's final pre-resolution consensus price (closing line) in addition to the resolution outcome. Section 4 develops why this multiplies statistical power.
- **Position netting and thesis grouping.** Fills are aggregated to episodes (open-to-flat per market) and episodes are grouped into theses — clusters of markets whose outcomes are mutually correlated (e.g., many contracts referencing one election). Effective sample size is computed at the thesis level; a wallet with 400 trades expressing one macro view has an effective n near 1.

3 · A taxonomy of participants

Before any ranking, each wallet is typed. Typing serves two purposes: it removes non-forecasting flow from the leaderboard population, and it routes distinct signal products to distinct consumers (informed-flow alerts and fade signals are valuable precisely because they are *not* sharp flow).

Type	Economic role	Discriminating features	Treatment
Sharp	Persistent positive-EV forecasting	Positive CLV at scale; sizing–edge correlation; timing lead; category specialisation; abstention elasticity	Leaderboard; rating published
Square (recreational)	Entertainment-driven negative-EV flow	Negative CLV; entries chase price moves; sizing correlates with recent results; event-driven activity bursts	Excluded from ranks; aggregate fade signal
Lucky monkey	Noise mistaken for skill	Strong short-window P&L; small or thesis-concentrated n; CLV ≈ 0; streak clustering	Heavily shrunk by the prior; not promoted
Market maker	Spread capture	Two-sided quoting in the same market; high order/cancel ratio; near-zero net directional exposure	Excluded; liquidity analytics
Arbitrageur	Pricing-inconsistency capture	Paired YES/NO or cross-market structures; near-perfect win rate with capped per-trade return	Excluded; market-efficiency analytics
Late-sweeper	Harvesting near-resolved markets	Entry prices clustered > 90¢; very short holding periods near resolution; severe left tail	Excluded from skill ranks

Type	Economic role	Discriminating features	Treatment
Likely-informed	Possible non-public information	Fresh wallets; large early correct positions in niche, insider-prone categories; extreme timing lead	Flagged distinctly; institutional alert product; never promoted as “skill”
Manipulator	Record fabrication / wash trading	Counterparty concentration; opposite-side cluster wallets; shared funding graph; profit vs. few repeat counterparties	Excluded; integrity action (§7)
Hedger	Offsetting external exposure	Systematic negative on-platform EV; positions opposing identifiable external interests	Excluded from ranks

Table 1 — Participant taxonomy. Typing precedes rating; only Sharp-eligible wallets enter the published leaderboard population.

Typing is implemented as a supervised classifier over the feature dictionary of Appendix A, trained on hand-labelled archetypes and audited quarterly. Wallets carry type posteriors, not hard labels; ambiguous wallets are withheld from ranking rather than forced into a class.

4 · Signals of persistent skill

4.1 · Closing-line value (CLV)

CLV is the primary skill signal. For a fill at price p in market m, with pre-resolution closing price c, the per-trade CLV is the log-odds improvement captured at entry:

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CLV = logit(c) − logit(p) where logit(x) = ln(x / (1−x))
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The logic: the closing line aggregates all information the market ever assembled; an entry that systematically beats it has extracted information ahead of consensus, independent of how individual resolutions break. Because CLV is observed on *every* trade rather than only at resolution, it converges on true skill roughly an order of magnitude faster than P&L — and empirically, trailing CLV is far more autocorrelated across windows than trailing ROI, which is the operational definition of a persistent signal. Wallets are summarised by size-weighted mean CLV, its t-statistic, and its stability across theses.

4.2 · Sizing discipline

Sizing behaviour is observable from the first trade and loads almost entirely on the persistent component. Two statistics dominate:

- **Conviction–edge correlation:** corr(stake_i, CLV_i) across a wallet’s fills. Skilled participants size like noisy Kelly bettors — larger when righter. Values near zero indicate flat or habit sizing; negative values are a red flag.
- **Result-reactivity:** the regression of stake_t on recent realised P&L. Disciplined wallets show no dependence; recreational wallets escalate after wins (house-money) or after losses (chasing). Either dependence predicts non-persistence.

4.3 · Timing lead

For each entry, the subsequent price path is summarised by the drift toward (or away from) the wallet’s side over horizons of 1h–72h. Leaders lead: their entries precede informed moves, yielding a positive lead distribution that is stable per wallet and robust across market regimes.

Followers and squares enter after moves, near local extremes. Lead/lag is also the cleanest separator between genuine sharps and likely-informed wallets (whose lead is extreme but episodic).

4.4 · Selection structure and abstention

Skill concentrates. A wallet's category concentration (Herfindahl over market types), odds-band consistency, and within-specialty edge jointly indicate a real process rather than ambient betting. Equally diagnostic is abstention elasticity: sharps' activity collapses when nothing is mispriced and spikes when dislocations appear; squares bet whenever there is content. A wallet that cannot *not* bet is, with high probability, not selecting.

5 · Signals of transience

The transient component dominates short horizons; its signatures predict both near-term direction (weakly) and long-term decay (strongly).

- **Tilt signatures.** Stake escalation after losses, shrinking inter-trade intervals within losing sessions, session-length drift into late-night hours, deposit-and-deploy-immediately patterns. These indicate that results are driving behaviour rather than the reverse; a wallet exhibiting tilt after a strong run is rated *down*, because the process that generated the record is visibly degrading.
- **Streak structure.** Result sequences are tested against the binomial null. Genuine skill produces boringly independent outcomes; clustering beyond chance usually indicates regime-riding — one thesis paying repeatedly — which predicts sharp reversal when the regime turns. Thesis-level P&L concentration quantifies the same risk.
- **Form vs. baseline.** Short-horizon expectations use a decayed performance estimate (half-life 60–90 days) shrunk toward the wallet's own long-run skill posterior — not toward zero. Hot streaks by low-CLV wallets revert hard; cold streaks by high-CLV wallets are deviations, not information.

6 · The Sharpness Rating model

6.1 · Hierarchical structure

Each Sharp-eligible wallet i carries a latent per-trade edge θ_i (in price-relative, log-odds units). The population prior is deliberately pessimistic — mass concentrated near zero with a thin positive tail — reflecting the empirical reality that most participants have no edge:

$$\theta_i \sim \pi(\theta \mid x_i) \quad \text{observed } CLV_{ij} \sim N(\theta_i \cdot d_j \, , \sigma^2_j)$$

where x_i is the wallet's behavioural covariate vector (sizing discipline, timing lead, selection structure, tilt indicators), d_j a decay weight on trade j, and σ^2_j a per-trade noise term scaled by market liquidity and thesis correlation. Two design choices carry the system:

- **Behaviour shifts the prior.** The covariates x_i translate and reshape $\pi(\cdot)$ before outcome evidence arrives. Two wallets with identical 100-trade records receive different posteriors if one exhibits Kelly-like sizing and timing lead while the other shows flat sizing and chase entries. This is where Sections 4–5 enter the mathematics rather than the marketing.

– **Decay handles edge erosion.** Exponential down-weighting (half-life a tunable 6–12 months for the persistent component) reflects that edges in adversarial markets decay as they are discovered — including via TailSharp itself.

6.2 · Conservative ranking

Published rank is determined not by the posterior mean but by a lower confidence bound:

$$\text{Rating}_i = Q_\alpha(\ \theta_i \mid \text{data}\ ) \quad \alpha \approx 0.10\text{–}0.20$$

Ranking by Q_α — a PAC-Bayes-motivated choice — automatically punishes small samples and thesis concentration, rewards consistency, and is far harder to game with variance: a manipulator can inflate a mean cheaply, but inflating a lower bound requires sustained, diversified, evidence-backed performance, which is indistinguishable from actually being good.

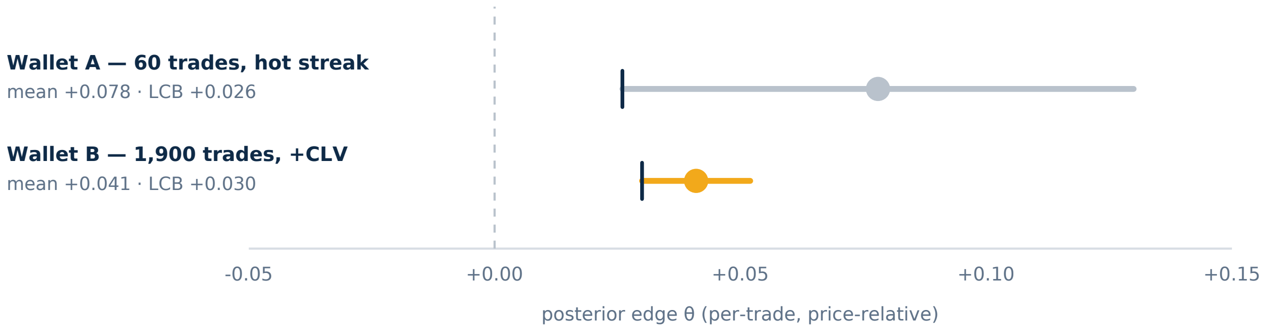

Figure 2 — Why the lower bound ranks. Wallet A's 60-trade streak has the higher posterior mean; Wallet B's 1,900-trade record has the higher lower confidence bound (tick marks) and ranks first.

The public output per wallet is the rating, $P(\theta_i > 0)$, effective sample size, specialty decomposition, and the behavioural evidence summary — a typed, auditable claim rather than a bare number.

7 · Manipulation resistance and wallet clustering

A rating worth following is a rating worth attacking. Three adversaries matter.

– **Wash-record builders** operate paired wallets on opposite sides, sacrificing one to decorate the other. Defences: counterparty-concentration statistics (share of profit earned against the k most frequent counterparties), opposite-side co-occurrence graphs, and the funding-graph clustering below. Profit that cannot be shown to come from the open market does not enter the rating.

– **Sybil splitters** spread activity across many wallets — sharps hiding from trackers, or manipulators running many small records hoping one runs hot. Wallets are clustered into entities using funding-source ancestry on-chain, withdrawal convergence, timing signatures (inter-trade interval distributions are surprisingly individual), gas/fee idiosyncrasies, and position-correlation fingerprints. Ratings attach to entities, not addresses; a split record is reassembled before it is scored. Entity resolution is probabilistic and conservative: merge only above high confidence.

– **Selective displayers** exploit any opt-in surface by showing winners and hiding losers. The on-chain setting removes this channel — coverage is total by construction — which is a structural integrity advantage over self-reported tipster records and is stated as such.

The clustering layer is also the system's durable moat: it is the component a competitor cannot replicate from public dashboards, and the component institutional data customers value most.

8 · Validation protocol

The rating earns publication only under walk-forward evidence. The protocol is fixed in advance:

- **Primary test:** at each month t, freeze ratings using only data ≤ t; measure rank correlation (Spearman / Kendall) between Rating_t and realised CLV over t+1 … t+6 for all rated entities. The rating must beat (i) trailing ROI ranks, (ii) trailing win-rate ranks, and (iii) a no-behaviour ablation of itself, out of sample, with the margin reported.
- **Calibration test:** published $P(\theta > 0)$ must be calibrated — of entities published at 70%, approximately 70% must exhibit positive forward CLV. Reliability diagrams are published quarterly.
- **Ablation suite:** each behavioural covariate family (sizing, timing, selection, tilt) is removed in turn and the predictive loss recorded, establishing which signals carry the rating and pruning those that do not survive contact with out-of-sample data.
- **Adversarial audit:** red-team wallets simulating wash, sybil, and streak-farming strategies are injected into the pipeline quarterly; the integrity layer's detection rates are tracked as a first-class metric.

All validation results, including failures, are versioned and published. The methodology's credibility is the product; the validation table is the sales document.

9 · Applications

- **The leaderboard:** Sharp-typed entities ranked by Rating (LCB), with evidence summaries, specialty decomposition, and effective sample sizes displayed inseparably from rank.
- **Position intelligence and alerts:** notifications when rated entities establish, extend, or exit positions — framed as market intelligence ("where is verified skill positioned?"), never as instructions to transact.
- **The Sharp Index:** a skill-weighted implied probability per market, pooling rated entities' positions via extremized log-odds aggregation with Rating-derived weights — a denoised consensus that can diverge informatively from the raw market price, licensable to media and institutions.
- **Flow taxonomies for institutions:** informed-flow flags, recreational-fade aggregates, and liquidity/maker analytics — byproducts of typing that are independently valuable and independent of any individual's transactions.

10 · Limitations, ethics, and positioning

- **Information product, strictly.** TailSharp analyses public on-chain activity. It does not execute, route, pre-fill, or individually advise transactions, and does not operate referral or affiliate arrangements with any wagering venue. Jurisdictional availability of underlying venues varies; nothing in the product constitutes encouragement to access services unlawful in the user's jurisdiction.

- **Replication gap.** Followers of public information face latency, liquidity, and price-impact costs; realised follower returns will generally undershoot leader records on thin books. The product therefore sells *intelligence about positioning*, not replication of returns, and says so.
- **Observer effects.** Measurement changes the measured: identified sharps may split wallets or alter timing, and copy pressure erodes the edges being ranked. The decay structure and entity resolution mitigate but do not eliminate this; the rating is a moving target by design.
- **Insider flow.** Likely-informed wallets are flagged, segregated from skill rankings, and never promoted; their existence is a property of the underlying markets that the methodology reports honestly rather than launders into "skill."
- **Responsible use.** Prediction-market participation involves risk of loss and, for some users, risk of harm. TailSharp's published materials carry responsible-play messaging, and tilt-pattern research is used to *de-rate* deteriorating wallets — never to target vulnerable users.

Past performance, however carefully decomposed, does not guarantee future results. The Sharpness Rating is a statistical estimate with published uncertainty, not a promise.

Appendix A · Feature dictionary (summary)

Family	Feature	Definition (per entity)
Skill / CLV	`clv_mean_w`	Size-weighted mean of logit(c) − logit(p) across fills
	`clv_t`	t-statistic of CLV against zero, thesis-clustered standard errors
Sizing	`kelly_corr`	corr(stake, per-trade CLV)
	`result_react`	β of stake_t on trailing realised P&L; (target ≈ 0)
	`stake_entropy`	Entropy of stake distribution; flags habit sizing
Timing	`lead_1h_72h`	Post-entry drift toward entity's side over 1–72h horizons
	`chase_score`	Share of entries within top decile of trailing price moves
Selection	`cat_hhi`	Herfindahl of activity over market categories
	`abstain_elast`	Activity elasticity to measured market dislocation
Transience	`tilt_index`	Composite: loss-escalation, interval-shrink, session drift
	`streak_z`	Runs-test z-score of result sequence vs binomial null
	`thesis_conc`	Share of P&L; attributable to largest correlated thesis
Structure	`two_sided`	Same-market opposite-side volume share (maker flag)
	`cp_conc`	Profit share vs k most frequent counterparties (wash flag)
Clustering	`fund_cluster`	Entity ID from funding-graph + timing-signature resolution

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